Task E3 — OTA Strategy Selection

Strategies:

Full Replacement: Entire fleet updated immediately. Lowest operational complexity but highest deployment risk if the model contains errors.

Canary Deployment: Only 10 trucks receive the update initially. Remaining 75 trucks are updated after successful validation. Same final cost as full replacement, but much lower operational risk.

Shadow Mode: New model runs alongside the existing model during evaluation before activation. Provides the highest confidence but introduces additional computational and operational complexity.

Given parameters:

Parameter	Value
Model File Size	280 KB (INT8 TFLite Model) = 0.2734MB
Update Frequency	Every 6 Weeks
Pilot Fleet Size	85 Trucks
M2M SIM Cost	₹0.10 / MB

OTA Deployment Cost Comparison:

Cost Metric	Full Replacement	Canary Deployment	Shadow Mode
Phase-1 / Initial Deployment	85 × 0.273 MB = **23.24 MB**	10 × 0.273 MB = **2.73 MB**	10 × 0.273 MB = **2.73 MB**
Phase-1 Cost	23.24 × ₹0.10 = **₹2.32**	2.73 × ₹0.10 = **₹0.27**	2.73 × ₹0.10 = **₹0.27**
Remaining Fleet Deployment	Not Applicable	75 × 0.273 MB = **20.51 MB**	75 × 0.273 MB = **20.51 MB**
Remaining Fleet Cost	Not Applicable	20.51 × ₹0.10 = **₹2.05**	20.51 × ₹0.10 = **₹2.05**
Total Trucks Updated	85	85	85

Total Bandwidth per Update Cycle	**23.24 MB**	**23.24 MB**	**23.24 MB**
Total OTA Cost per Update Cycle	**₹ 2.32**	**₹ 2.32**	**₹ 2.32**

Evaluation against FreightBridge requirements

1. Full Replacement

Advantages

- Fastest deployment.
- Simple to implement.
- Easiest operational process.

Disadvantages

- Any faulty model affects all 85 trucks immediately.
- Cold-chain pharmaceutical cargo may be exposed to missed detections.
- Large operational risk.

Verdict

- Not recommended for safety-critical deployments.

2. Canary Deployment

Advantages

- Limits risk to only 10 trucks initially.
- Real-world validation before fleet-wide rollout.
- Easy rollback if model performance degrades.
- Well suited for intermittent rural connectivity because deployment occurs in stages.

Disadvantages

- Slightly slower rollout.
- Requires monitoring during validation period.

Verdict

- Strong candidate.

3. Shadow Mode

Advantages

- Highest validation confidence.
- Existing model continues protecting cargo.
- Allows comparison between old and new models.

Disadvantages

- Higher operational complexity.
- Requires running two models simultaneously.
- Additional CPU, memory, and energy consumption on Raspberry Pi.

Verdict

- Technically strongest but introduces unnecessary complexity for a lightweight fleet deployment.

Recommended OTA Strategy

Canary Deployment: Justification

FreightBridge operates a cold-chain logistics fleet carrying temperature-sensitive pharmaceutical products. Any faulty model update could result in missed anomaly detection and potential cargo spoilage.

Canary deployment minimizes this risk by:

1. Updating only **10 trucks initially**.
2. Validating model behavior under real operating conditions.
3. Detecting issues before fleet-wide rollout.
4. Supporting rapid rollback if problems are observed.
5. Reducing operational risk while maintaining the same bandwidth cost (₹2.32 per update cycle).

Why Not Full Replacement?

- High fleet-wide risk.
- Any defective model impacts all trucks simultaneously.
- Unsuitable for a safety-critical cold-chain system.

Why Not Shadow Mode?

- Requires dual-model execution.
- Increases computational and operational complexity.
- Additional validation overhead provides limited benefit for a small 280 KB MLP model.

Final Answer

Item	Selected Strategy
Recommended OTA Method	**Canary Deployment**
Initial Deployment Group	**10 Trucks**
Fleet Rollout	Remaining 75 Trucks after validation
Bandwidth per Update Cycle	**23.24 MB**
Update Cost per Cycle	**₹ 2.32**
Primary Reason	Best balance of safety, reliability, and rural fleet deployment practicality
Full Replacement	Rejected due to fleet-wide update risk
Shadow Mode	Rejected due to unnecessary complexity and higher resource consumption